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$$\begin{aligned} & \int \sqrt{1 - \cos x} dx \\ &= \int \sqrt{2 \sin^2 \left(\frac{x}{2} \right)} dx \\ &= \pm \sqrt{2} \int \sin \left(\frac{x}{2} \right) dx \\ \text{Stel } t &= \frac{x}{2} \Rightarrow dt = \frac{dx}{2} \\ &= \pm 2\sqrt{2} \int \sin(t) dt \\ &= \pm 2\sqrt{2} \cos(t) + C \\ &= \pm 2\sqrt{2} \cos \left(\frac{x}{2} \right) + C \end{aligned}$$

References